

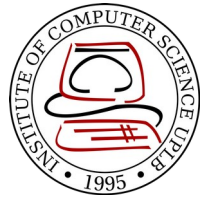
CMSC 137

Data Communications and Networking

Lecture Module
10 – Transmission Media



University of the Philippines
LOS BAÑOS



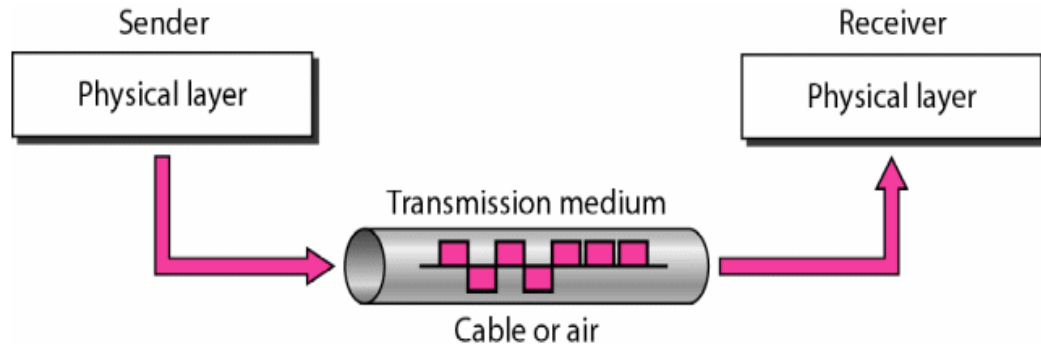
Module 10: Transmission Media

Rozano S. Maniaol
Lecturer

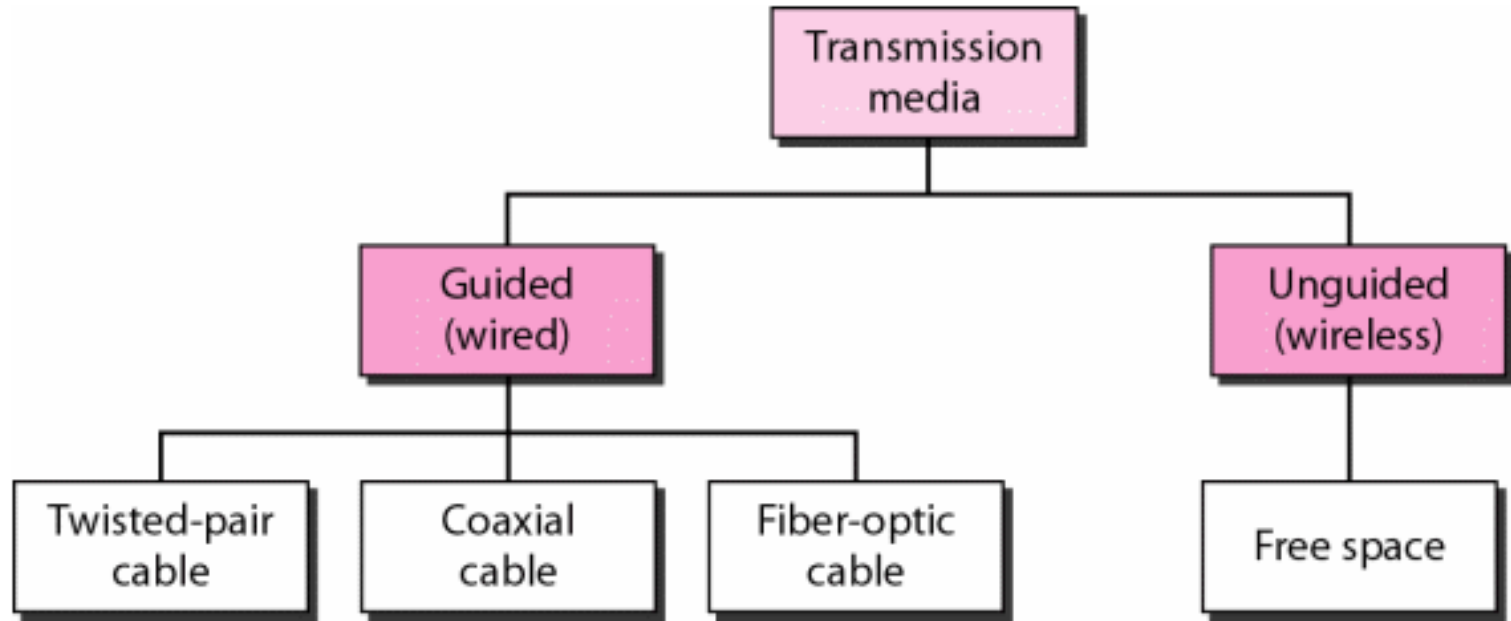
Introduction

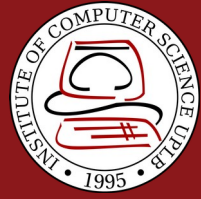


- Layer 0
- Anything that can carry information from a source to destination
- **Electromagnetic energy** – a combination of electric and magnetic fields vibrating in relation to each other: power, radio waves, infrared light, etc.
- Portions of electromagnetic spectrum

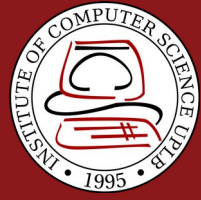


Classification of Transmission Media





Wired Media



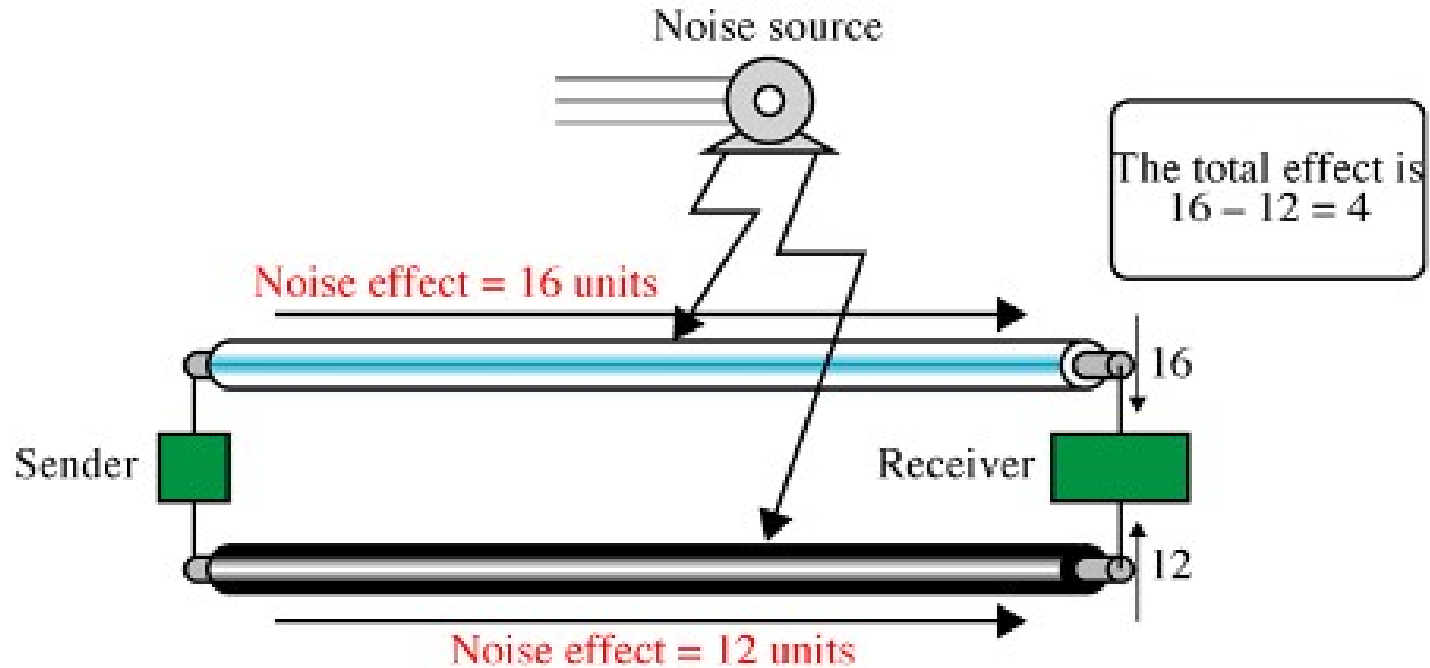
Twisted Pair Cable

Twisted Pair Cable

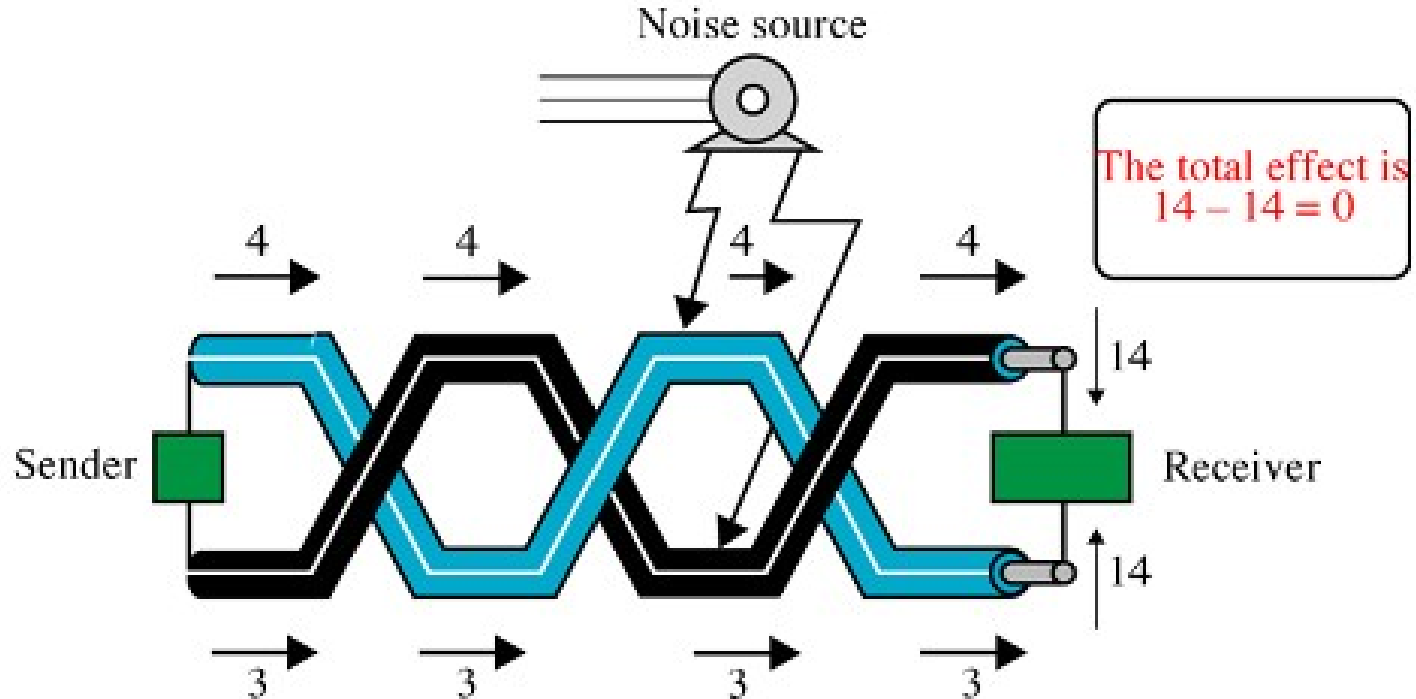


- Consists of 2 conductors, each with its own plastic insulation twisted together
- One wire is used to carry **signals** to the receiver, another is for **ground reference**
- If the two wires are parallel, effect of unwanted signals is not the same since they are located at different distance relative to the noise
- Applications: telephone lines, DSL, LANs

Effect of Noise on Parallel Lines



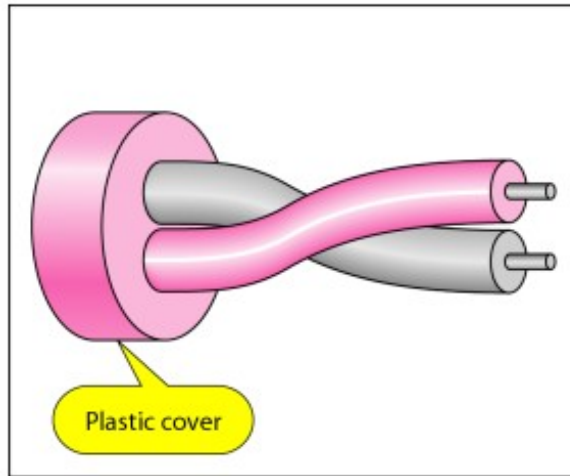
Effect of Noise on Twisted Pairs



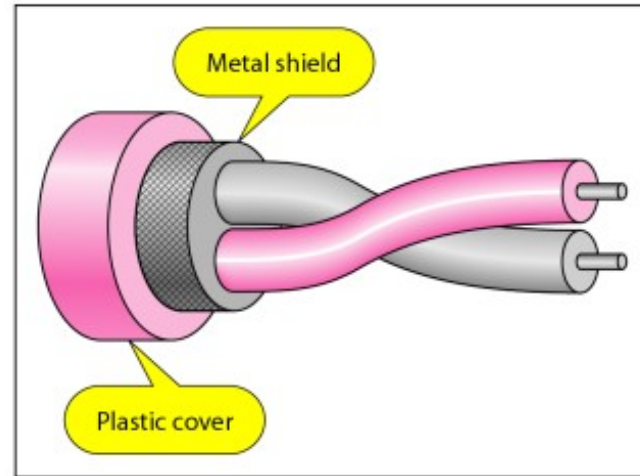
Shielded vs Unshielded Twisted Pair



- **Unshielded Twisted-Pair** (UTP) - most common twisted-pair cable used
- **Shielded Twisted-Pair** (STP) – produced by IBM; has metal foil or braided-mesh covering

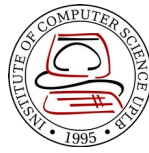


a. UTP



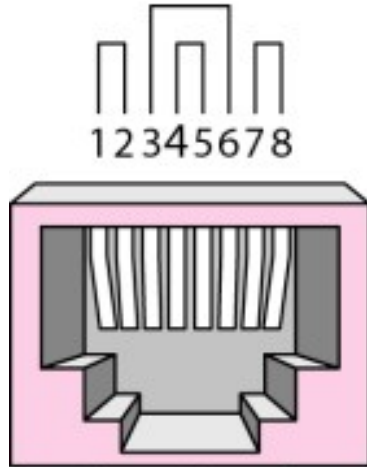
b. STP

Categories of Copper Cables

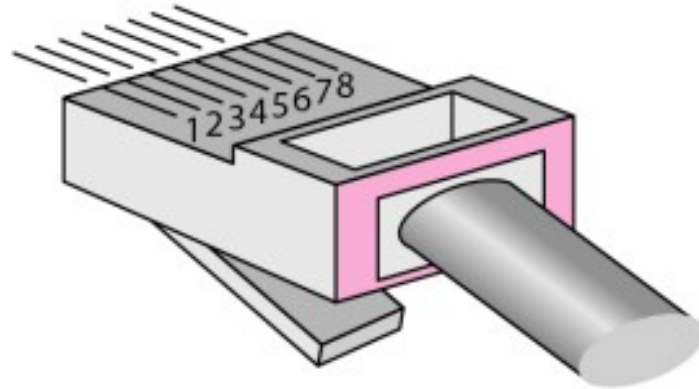


Category	Data Rate	Max Length	Application
CAT 1	Up to 1 Mbps	-	Old telephone cables
CAT 2	Up to 4 Mbps	-	Token Ring networks
CAT 3	Up to 10 Mbps	100m	Token Ring & 10BASE-T Ethernet
CAT 4	Up to 16 Mbps	100m	Token Ring networks
CAT 5	Up to 100 Mbps	100m	Ethernet, Fast Ethernet, Token Ring
CAT 5e	Up to 1 Gbps	100m	Ethernet, Fast Ethernet, Gigabit Ethernet
CAT 6	Up to 10 Gbps	100m	Gigabit Ethernet, 10G Ethernet (55m)
CAT 6a	Up to 10 Gbps	100m	Gigabit Ethernet, 10G Ethernet (55m)
CAT 7	Up to 10 Gbps	100m	Gigabit Ethernet, 10G Ethernet (100m)

UTP Connector



RJ-45 Female



RJ-45 Male

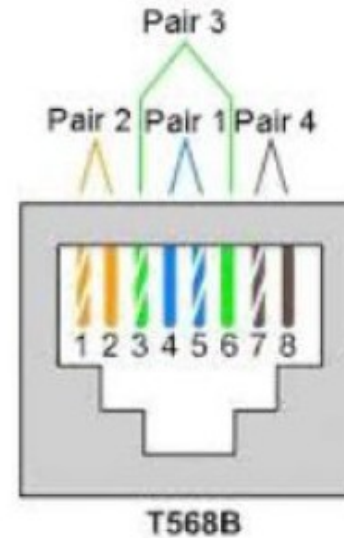
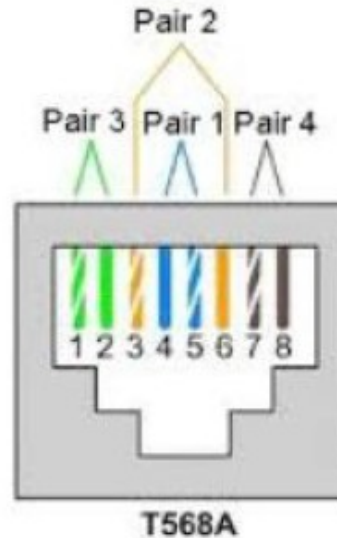
Legend:

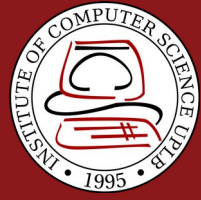
RJ - Registered Jack

UTP Color Coding Standard



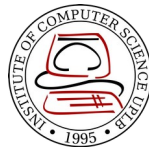
Pin	Assignment
1	Transmit +
2	Transmit -
3	Receive +
4	Unused
5	Unused
6	Receive -
7	Unused
8	Unused



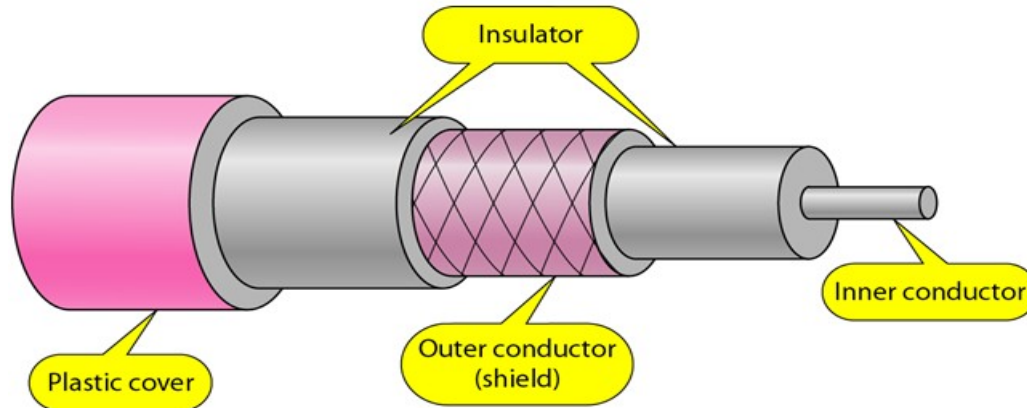


Coaxial Cables

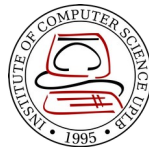
Coaxial Cables



- Has a central core conductor of solid or stranded wire enclosed in an insulating sheath, which is in turn, encased in an outer conductor of metal foil, braid or combination of the two
- Can carry signals of higher frequencies than twisted pair
- Applications: telephone lines, cable, LANs



Categories



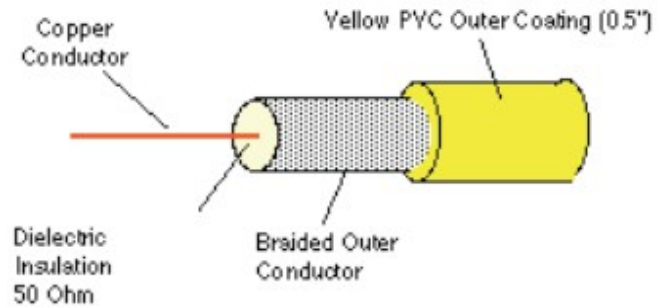
- Thicknet (10BASE-5) – uses RG-11 to transmit 10 Mbps with a range of 5000m
- Thinnet (10BASE-2) – uses RG-58 with BNC connectors to transmit data at 10 Mbps with a range of 185m

<i>Category</i>	<i>Impedance</i>	<i>Use</i>
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

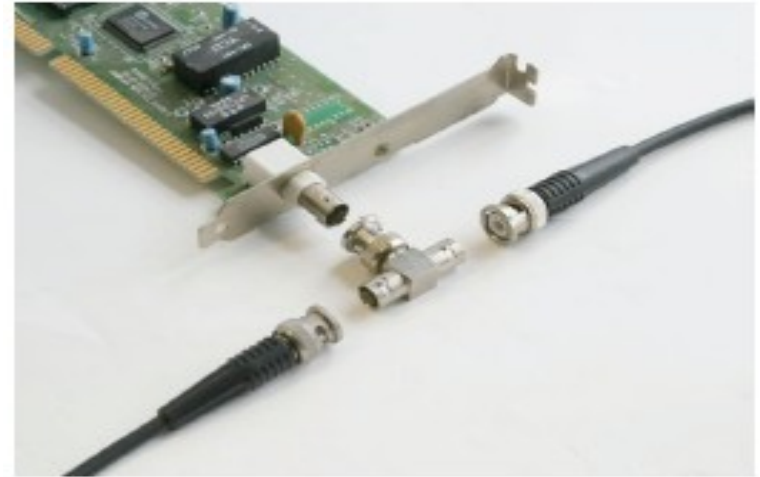
Legend:

RG – Radio Government

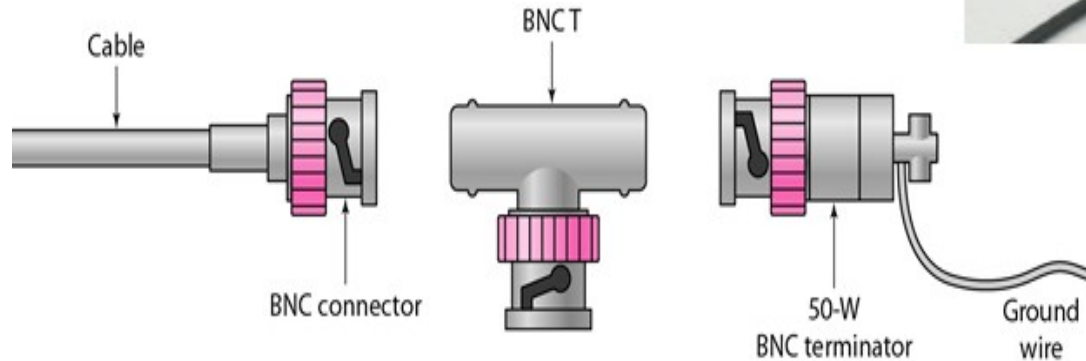
RG-11

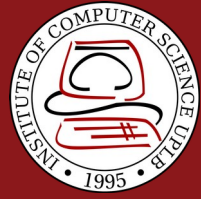


Bayonet-Neill-Concelman (BNC)



Source: nsr.org



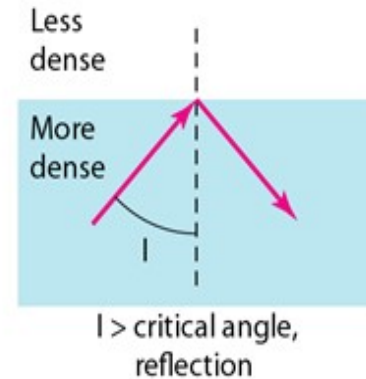
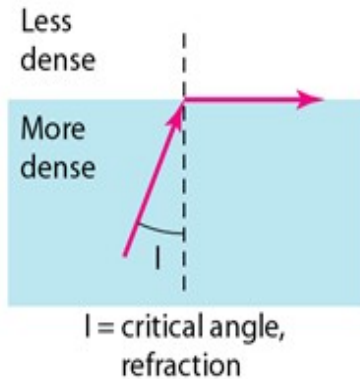
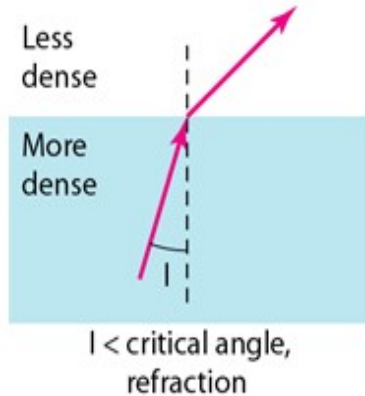


Fiber Optic Cables

Fiber Optic Cable



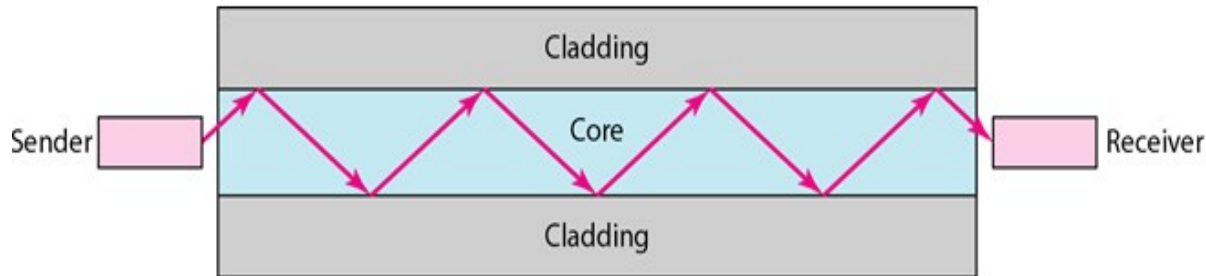
- Made of glass or plastic, transmit signals in the form of light
- Nature of light
 - Light travels in a straight line as long as it is moving through a single uniform substance
 - If a ray of light traveling through one substance suddenly enters another substance (of a different density), the ray changes its direction



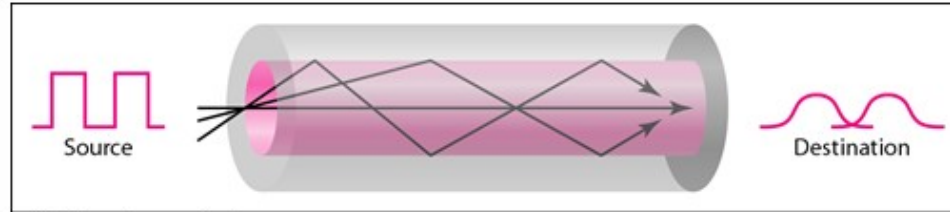
Fiber Optic Cable



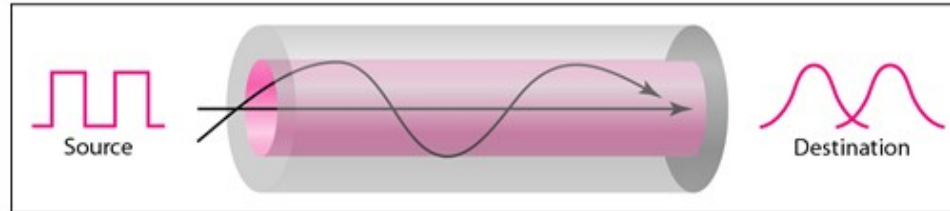
- Uses reflection to transmit
- Glass or plastic core is surrounded by a cladding made of less dense glass or plastic



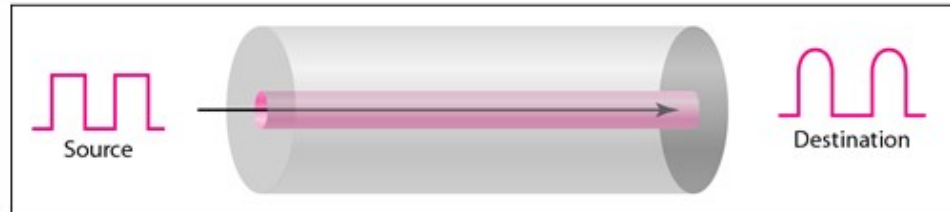
Propagation Modes



a. Multimode, step index

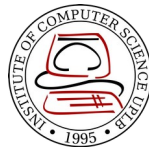


b. Multimode, graded index

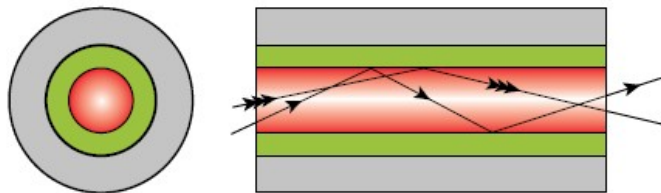
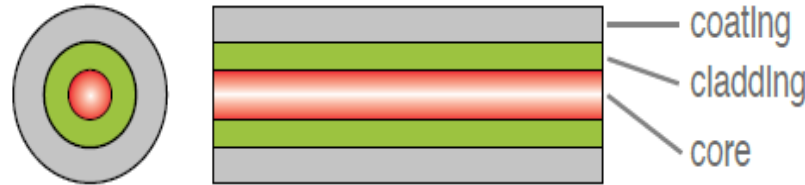


c. Single mode

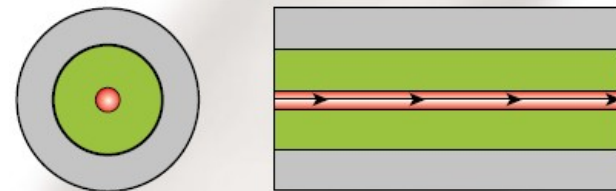
Fiber Types



Type	Core (μm)	Cladding (μm)	Mode
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

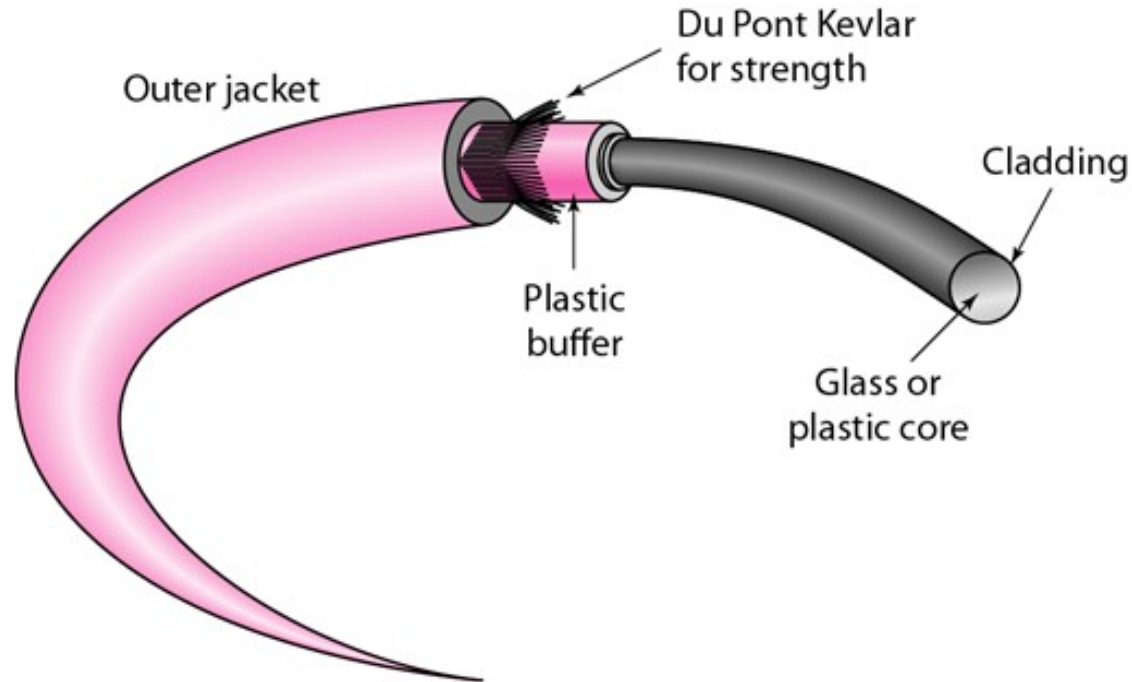


Multimode



Single mode

Fiber Construction



Fiber Packages

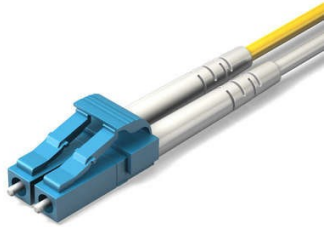
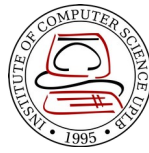


Indoor Fiber Packages



Outdoor Fiber Packages

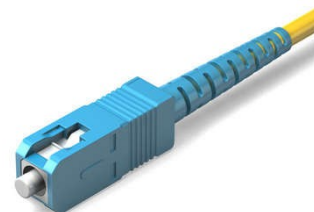
Fiber Optic Connectors samples



LC connector



MTRJ connector



SC connector



ST connector

<https://www.cablestogo.com/learning/connector-guides/fiber-networking>

Legend:

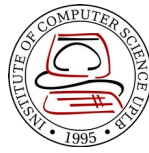
LC – Lucent Connector

MTRJ – mechanical transfer registered jack

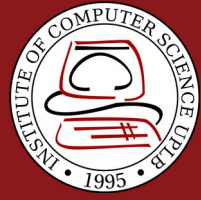
SC – subscriber channel

ST – straight tip

Fiber Optic Cables



- Advantages
 - Higher bandwidth
 - Less signal attenuation
 - Immunity to electromagnetic interference
 - Resistance to corrosive materials
 - Lightweight
 - Greater immunity to tapping
- Disadvantages
 - Installation and maintenance
 - Unidirectional light propagation
 - Cost

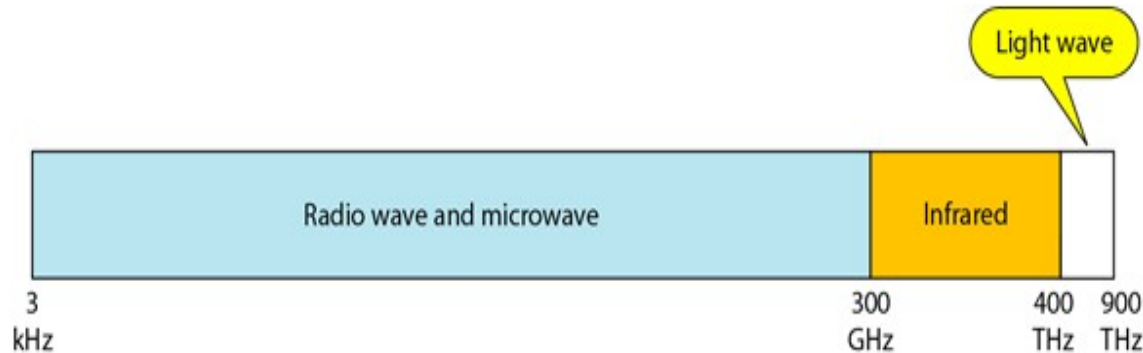


Wireless Media

Unguided Media



- Transmit without a physical conductor
- Aka wireless communication
- Signals broadcasted through free space and available to anyone with a receiver



Electromagnetic spectrum for wireless communication

Propagation Mode



- Ground Propagation
 - Radio waves travel through the lowest portion of the atmosphere, hugging the earth
 - The low frequency signal follow the curvature of the planet
 - Distance depends on the amount of the power
- Sky Propagation
 - Higher frequency radio radiate upward into the ionosphere where they are reflected back to the earth
 - Sky propagation allow for greater distance with lower power output
- Line of Sight Propagation
 - Very high frequency signals are transmitted in straight lines directly from antenna to antenna

Propagation Mode



Ionosphere



Ground propagation
(below 2 MHz)

Ionosphere



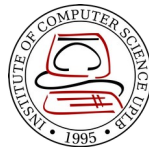
Sky propagation
(2–30 MHz)

Ionosphere



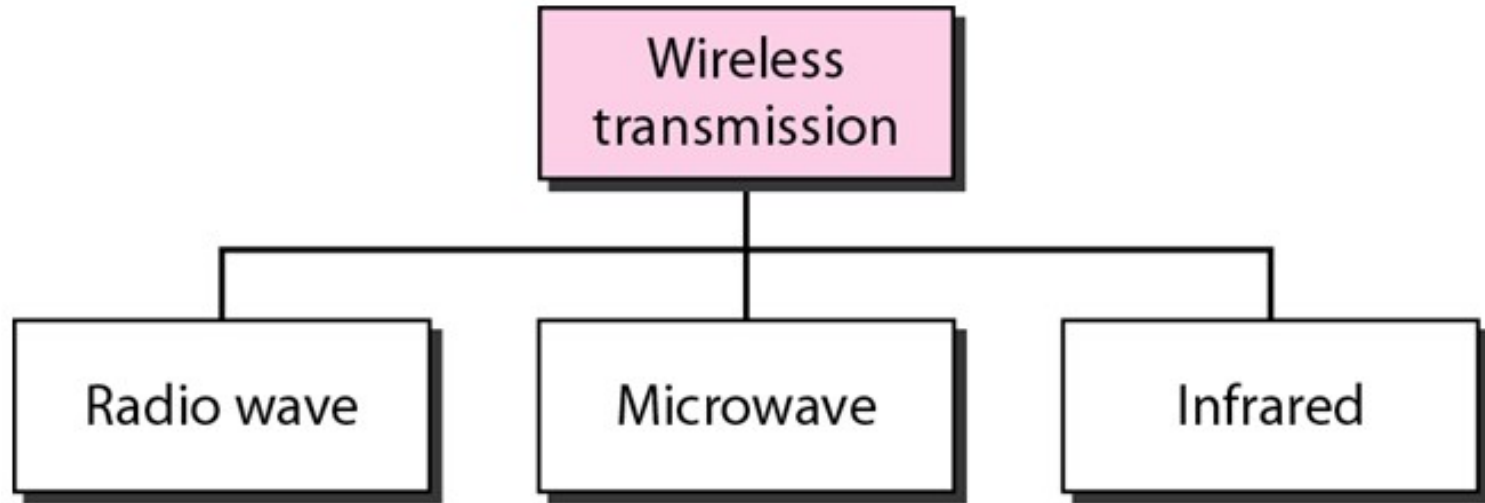
Line-of-sight propagation
(above 30 MHz)

Bands

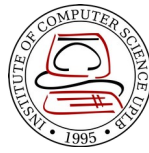


<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30–300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite

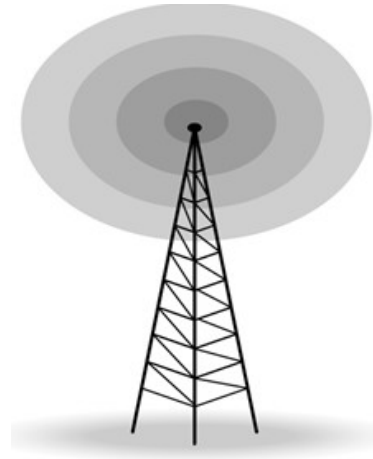
Wireless Transmission Group



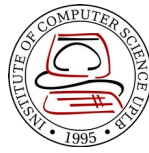
Radio Waves



- Frequency range of 3 kHz to 1 GHz
- **Omnidirectional** – signals propagate in all directions
- Signals can penetrate walls
- Susceptible to interference by another antenna that sends signals using the same frequency or band
- Applications: multicasting

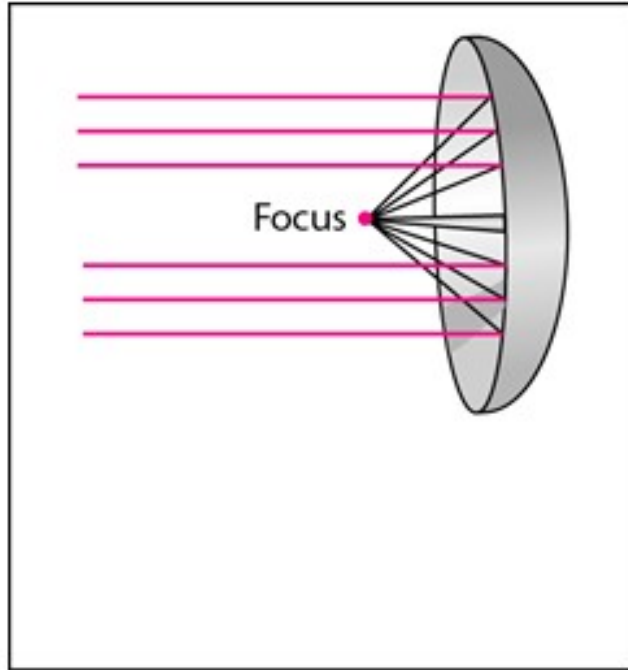
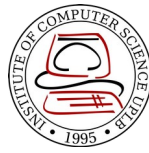


Microwave

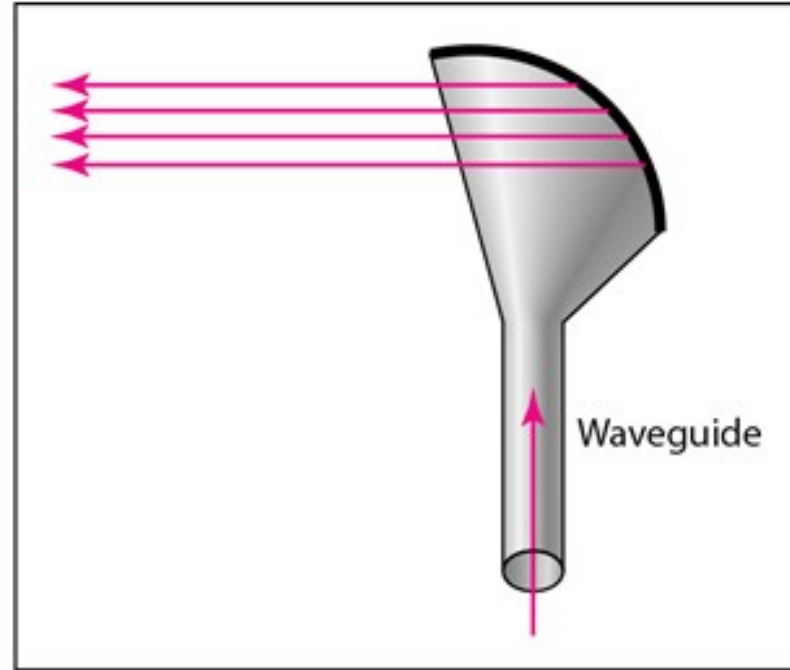


- Frequency range from 1 to 300 GHz
- **Unidirectional** – sending and receiving needs to be aligned
- Line-of-sight propagation
- Signals cannot penetrate walls
- Wider sub bands and higher data rate
- Applications: unicast, cellular networks, satellite networks, wireless LANs

Unidirectional Antenna

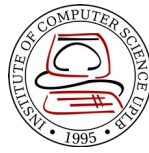


a. Dish antenna



b. Horn antenna

Infrared



- Frequency range from 300 GHz to 400 THz
- Short range communication
- Signals cannot penetrate walls
- Applications: IrDA (Infrared Data Association) port, communication between devices
- Line-of-sight propagation

References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.